
THE DOPAMINERGIC TROPHIC CASCADE FOLLOWING SARS-CoV-2 INFECTION: A SYSTEMS VIEW OF NEUROLOGICAL LONG COVID

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ABSTRACT

Coming Soon

Introduction

Neurological post-acute sequelae of SARS-CoV-2 infection (Neurological PASC) is characterized by a recurring constellation of cognitive dysfunction, fatigue, post-exertional malaise, autonomic disturbance, sleep disruption, affective symptoms, and motor impairment. Considerable progress has been made identifying biological abnormalities associated with this syndrome, including viral persistence, neuroinflammation, endothelial dysfunction, microvascular injury, metabolic impairment, astrocytic pathology, and immune dysregulation (Greene et al., 2024). These findings have substantially advanced understanding of the disease, yet they remain largely descriptive. They identify processes occurring in Neurological PASC without fully explaining why these processes repeatedly converge upon Long COVID phenotype or why symptoms emerge and evolve in the characteristic manner observed clinically (Davis et al., 2021).

This paper proposes that Neurological PASC represents a distinct neurological disorder whose defining features emerge from a common systems-level mechanism: a biological cascade in brain systems initiated when dopaminergic regulatory output falls below a functional threshold required to maintain downstream neural systems. Rather than treating the symptom profiles of Long COVID as independent consequences of the SARS-CoV-2 virus itself, this framework hypothesizes a cascading sequence where they are expected outcomes of an integrated system failure termed the *Dopaminergic Trophic Cascade* (DTC).

This paper is not a hypothesis about a unique mechanism by which SARS-CoV-2 produces neurological disease. Rather, it proposes a general systems architecture governing dopaminergic failure. SARS-CoV-2 serves as the principal clinical example because of its epidemiological importance and the difficulty of explaining its neurological sequelae through existing frameworks (Greenhalgh, Sivan, Perlowski, & Nikolich, 2024; Kenny, Townsend, Savinelli, & Mallon, 2023), but the cascade itself is defined independently of the initiating trigger. The virus may trigger the cascade because it infects support cells and causes dysfunction, but once that dysfunction reduces output past what is required for dependent systems, the threshold is crossed and the effects follow according to brain architecture.

The trophic cascade concept originates in ecology. In a trophic cascade, the removal of a keystone regulatory species produces disproportionate and cascading dysfunction across the entire ecosystem because so many dependent species rely on that regulatory presence. Remove the keystone, and the structure collapses in ways no single-species analysis would have predicted. In the brain, a parallel architecture exists. Through regulation, dopaminergic neurons function as hubs governing motor control, cognition, motivation, autonomic regulation, and sleep-wake dynamics (Dzirasa et al., 2006). Their functional integrity depends, in turn, on a support architecture: astrocytes. These support cells supply metabolic substrates, regulate extracellular glutamate and potassium, and maintain the microenvironment required for sustained dopaminergic output. Once compromised, the support infrastructure affects the regulatory capacity of

the dopaminergic system. When the capacity required to support other systems falls below a critical threshold, the Dopaminergic Trophic Cascade is initiated. What follows is not determined by the identity of the initiating trigger, but by the architecture of the dopaminergic system itself. The resulting process is systemic, hierarchical, self-amplifying, and capable of generating self-reinforcing feedback loops.

The initiating trigger and the cascade are separate. The initiating trigger reduces dopaminergic functional output. The cascade describes the system's behavior after output falls below that required by dependent systems. The identity of the trigger is therefore contingent, whereas the cascade is a property of the dopaminergic system itself. Viral infection, endocrine withdrawal, or other initiating triggers capable of sufficiently reducing dopaminergic functional output are predicted to converge on the same downstream architecture; the cascade fails the same way because it is a property of the whole neurotransmitter system, not the trigger. All systems operate with metabolic costs and occasionally require the outputs of other systems as inputs. The questions this paper asks concern these inputs and outputs. Specifically, is there a system capable of affecting multiple downstream systems and thereby producing the characteristic neurological syndrome of Long COVID? Recognition of the dependence of other systems on the dopaminergic system suggested that dysfunction within this system could have far-reaching consequences throughout the brain that no other neurotransmitter system could equal.

Current literature cannot explain why symptoms cluster as they do within individuals and populations, why onset is systematically delayed after acute illness, why exertion produces a disproportionate and delayed collapse, or why the symptom profile so closely resembles established disorders of dopaminergic dysfunction including Parkinson's disease and ADHD (Tsampasian et al., 2023). Anti-inflammatory interventions have shown limited and inconsistent efficacy (Knopman et al., 2026). Plasma biomarkers of neuronal injury and neuroinflammation are not significantly elevated in Long COVID patients compared to recovered controls (Omdal, Lenning, Jonsson, Kvaløy, Di Molfetta, et al., 2026)—a result that challenges the inflammation-based causal account while leaving the clinical reality of persistent symptoms unexplained.

Many initiating triggers may reduce dopaminergic functional output sufficiently to initiate the cascade, but the implications of those triggers beyond Neurological PASC are outside the scope of this paper. The threshold-crossing event that initiates the cascade is separate from the resulting systemic consequences that propagate throughout the brain's ecosystem. Direct viral infection of astrocytic support cells is one initiating trigger: the virus gains direct access to the support infrastructure and degrades its homeostatic function (Crunfli et al., 2020). Direct infection of dopaminergic neurons can further compromise their function, reducing dopamine synthesis and release (L. Yang et al., 2024). The dopamine synthetic pathway itself, from tyrosine through TH-mediated hydroxylation, has been specifically identified as vulnerable to SARS-CoV-2-mediated disruption (Nataf, 2020). Hormone-driven trophic withdrawal—as occurs during perimenopause with the erratic decline and eventual loss of estradiol—constitutes another potential initiating trigger that reduces dopaminergic synthesis capacity from within (Jacobs & D'Esposito, 2011; Shah et al., 2025). These are not competing explanations. Each is capable of independently initiating the same cascade if dopaminergic functional output is reduced below that required by dependent systems. The initiating trigger is the event that produces the reduction in output. The cascade is the subject of this hypothesis.

The field has not failed to generate data. What is missing is the right level of explanation. Existing models explain important individual observations, but no single framework explains, simultaneously, the delayed onset of symptoms, their characteristic clustering, polygenic susceptibility, female predominance, profound post-exertional malaise, normal structural biomarkers, and abnormal functional neuroimaging (Douaud et al., 2022; Guedj et al., 2021). The question is therefore not simply what SARS-CoV-2 does to the brain, but how one regulatory architecture can organize these observations into a coherent neurological syndrome. The model does not propose that dopamine directly mediates every neurological manifestation of PASC. Rather, it proposes that failure of the dopaminergic system impairs multiple downstream physiological systems into a coherent syndrome when taken together.

The Trophic Cascade Model

The cascade is organized as a sequential progression. The initiating trigger is external to the cascade and does not alter its internal organization. Different initiating triggers may initiate the cascade, but once initiated, progression is governed by the architecture of the dopaminergic system rather than by the identity of the initiating trigger. The model maps this vulnerability through four steps, beginning with an external initiating event, followed by threshold crossing, dopaminergic pathway dysfunction, and downstream effects that propagate through excitatory-metabolic overload and lymphatic clearance failure.

A: External Initiating Trigger

The sequence opens with an initiating trigger affecting the host's central nervous system. This step is external to the cascade proper. The DTC model posits that the identity of the initiating trigger is functionally irrelevant once dopaminergic functional output falls below that required by dependent systems. In Neurological PASC, viral infection is the principal clinical example, but it is not the only possible initiating trigger. Other initiating triggers capable of producing the same reduction in dopaminergic functional output are expected to initiate the same cascade because the cascade is a property of the dopaminergic system rather than of the initiating trigger.

Direct viral infection of astrocytic support cells is one initiating trigger: the virus degrades its homeostatic function and affects astrocytic support infrastructure (Crunfli et al., 2020). Infection of astrocytes by SARS-CoV-2 variants Delta and Omicron induces distinct transcriptomic signatures affecting neuronal system pathways, metabolism, and cytokine signaling (Bhide et al., 2025). Direct infection of dopaminergic neurons is another initiating trigger that can further compromise function, reducing dopamine synthesis and release (L. Yang et al., 2024). SARS-CoV-2 infection of human iPSC-derived dopaminergic neurons has been shown to hamper dopamine production (capelletti2023). The dopamine synthetic pathway itself, from tyrosine through TH-mediated hydroxylation, has been specifically identified as vulnerable to SARS-CoV-2-mediated disruption (Nataf, 2020). Hormones constitutes a third initiating trigger as occurs during perimenopause with the decline and eventual loss of estradiol (Jacobs & D'Esposito, 2011; Shah et al., 2025). These triggers are not competing explanations; They are functionally equivalent triggers that push the dopaminergic system toward the cascade threshold.

Step 1: Decreased Functional Output

The cascade itself begins at Step 1 only after the initiating trigger has sufficiently caused dopaminergic functional output to drop below what is required by dependent systems' normal function. Whatever condition is responsible for the decreased functional output is not a property of the cascade. It is the condition that initiated it. Once that threshold has been crossed, the dopaminergic system can no longer sustain its own normal function or the normal functioning of dependent, downstream systems without dysfunction. From this point onward, the architecture and wiring of the brain in concert with an individual's genetic profile influences the direction and degree of the cascade.

Step 2: Dopaminergic Pathway Dysfunction

Once threshold crossing has initiated the Dopaminergic Trophic Cascade, it immediately begins to affect the primary dopaminergic pathways resulting in dysfunction (ahlersdannen2020). The role of the dopaminergic system across neurological domains means that even partial dopaminergic dysfunction produces coherent neurological symptoms across multiple dependent systems. The specific symptom profile depends on which pathways are most affected and on individual differences in dopaminergic functional output and compensatory mechanisms.

Which pathways fail first, and how severely, is partially determined by the structural parameters of the pathways themselves, not solely by the magnitude of the Step 1 functional drop. These parameters are genetically encoded. Dopamine receptor density and subtype composition vary across individuals and are partially heritable. Variants in genes governing dopamine transporter expression shape how rapidly dopamine is cleared from the synapse. Genotypes governing G-protein signaling downstream of dopamine receptors, including RGS4 and RGS9, which are established modulators of dopaminergic control in striatal pathways, determine whether a given dopaminergic signal can be efficiently transduced and terminated, or whether its loss produces disorderly excitatory rebound (ahlersdannen2020). Genome-wide analyses of ADHD have identified 27 risk loci, including genes enriched in midbrain dopaminergic neurons, highlighting the polygenic architecture of dopaminergic pathway vulnerability (Demontis et al., 2023).

The nigrostriatal pathway, running from substantia nigra to striatum, supports movement initiation, coordination, and sustained motor output (Braun, Graham, Schaefer, Vorhees, & Williams, 2012). Partial dysfunction in this pathway produces motor fatigue, reduced physical endurance, subtle slowing of movements, and an effortful quality to physical activity that resembles mild Parkinsonian features (Brundin, Nath, & Beckham, 2020). Intracellular dopamine levels in Parkinson's disease, when adjusted for cell number and VMAT2 levels, are not as profoundly depleted as tissue concentrations suggest, implicating dopamine-mediated toxicity in disease progression (sacknerbernstein2021).

The mesocortical pathway projects from the ventral tegmental area to prefrontal cortex and modulates executive functions including working memory, planning, and cognitive flexibility (Eskenazi et al., 2021). Dysfunction in this pathway manifests as brain fog, slowed cognitive processing, difficulty initiating or sustaining complex mental tasks, impaired working memory, and reduced ability to shift between cognitive demands.

The mesolimbic pathway connects the ventral tegmental area to nucleus accumbens and other limbic structures, modulating motivation, reward processing, and affect (Eskenazi et al., 2021). Impairment produces anhedonia, loss of

drive, impaired motivation to initiate activities, reduced pleasure in previously enjoyed activities, and mood disturbances consistent with mesolimbic dopamine pathway dysfunction.

Dopaminergic neurons in the hypothalamus influence autonomic output, endocrine regulation, temperature control, and sleep-wake interactions (Dzirasa et al., 2006). Disturbance in these pathways produces postural tachycardia syndrome, orthostatic intolerance, temperature dysregulation, metabolic changes, and disrupted sleep architecture (Quan et al., 2025). Dopaminergic dysfunction impairs sleep quality and particularly reduces slow-wave sleep (Dzirasa et al., 2006), which has consequences for the next cascade level.

Dopamine does not selectively regulate one pathway over another. All dopaminergic pathways are affected when functional output falls below that required by dependent systems. However, the relative severity of dysfunction across these pathways varies between individuals based on the genetic architecture governing receptor density, transporter efficiency, and signaling dynamics in each pathway. The apparent heterogeneity of Neurological PASC is not evidence of multiple neurological diseases. It is the expected consequence of a single regulatory architecture failing in individuals with different genetic architectures. The particular combination and relative severity of symptoms reflect which pathways, in that individual's genetic profile, carry the least structural capacity as dopaminergic functional output declines.

Step 3: Downstream Effects

At Step 3, the dopamine pathway failures branch into two routes of systemic physiological dysfunction. These routes are mechanically linked and mutually reinforcing, converting a centralized neurotransmitter deficiency into chronic dysregulation.

Step 3A: Excitatory-Metabolic Overload

Dopaminergic and glutamatergic pathways are tightly coupled in fronto-striatal and limbic networks (**mancini2023**). Dopamine functions as a critical modulator of glutamatergic excitation, providing regulatory braking on excitatory drive. When dopaminergic tone collapses at Step 2, this excitation-inhibition balance is lost. Pathways are forced into a state of chronic glutamatergic overdrive to maintain output. The consequence is unbuffered cellular energy demand, escalating metabolic stress, and ultimately mitochondrial failure. The clinical expression is the characteristic burning cognitive fatigue that follows even minor mental effort (Rudroff, 2024).

The SARS-CoV-2 spike protein subunit S1 has been shown to induce endolysosome dysfunction and cellular senescence in primary human astrocytes through TLR7-dependent p38 MAPK signaling, suggesting a mechanism by which persistent viral antigens drive chronic neuroinflammation and metabolic dysfunction (Hasler et al., 2025). Furthermore, SARS-CoV-2 ORF3a expression induces the upregulation of Sur1-regulated ion channels in astrocytes, leading to intracellular calcium influx, apoptosis, and necrosis through both NF- κ B-dependent and independent proinflammatory pathways (Gerzanich et al., 2025). These astrocytic pathologies may compound the metabolic burden at Step 3A by impairing the support infrastructure required for sustained dopaminergic output and glutamate homeostasis.

Variants in RGS proteins, particularly RGS4 and RGS9, regulate the termination of G-protein signaling downstream of both dopamine and glutamate receptors in striatal pathways (**lernerrgs2012**). When an individual carries variants that reduce RGS efficiency, the excitatory rebound following dopaminergic withdrawal is sharper and more sustained. The mitochondrial thioredoxin system, governed in part by TXNRD2 and related redox regulators, maintains the cellular antioxidant environment required for sustained ATP production under metabolic stress (Nayeri et al., 2022). Individuals carrying constrained mitochondrial redox capacity arrive at energy failure sooner and recover from it more slowly.

Monoamine oxidase A activity further modulates this tier. Low-activity MAOA variants reduce catecholamine degradation rates (Gizer, Ficks, & Waldman, 2009). In the context of the Step 1 functional loss, the reduced clearance of norepinephrine and the broader catecholamine environment, combined with the glutamatergic overdrive state, produces prolonged excitatory tone that compounds the metabolic burden. The net effect is a longer and more severe Step 3A state for individuals with low-activity MAOA variants, because the excitatory environment cannot be efficiently reset between episodes of demand.

The Step 3A pathway creates its own feedback loop. Glutamatergic overdrive produces metabolic waste. That waste accumulates in the same dopaminergically innervated regions whose dysfunction initiated the cascade. The accumulation deepens cellular stress, further impairing the already compromised dopaminergic output. The excitatory environment intensifies, metabolic demand escalates, and the system cannot locate a stable state. This is not a transient response to exertion. It is a self-perpetuating metabolic trap.

Step 3B: Glymphatic Clearance Failure

The mechanism through which dopaminergic collapse translates into glymphatic clearance failure operates through a direct neurochemical dependency. Norepinephrine is the primary neuromodulator governing slow-wave sleep architecture and is the direct driver of the infraslow oscillations that mechanically propel astrocytic glymphatic fluid flow through the perivascular space (Hablitz, Vinitsky, Sun, et al., 2019; Xie et al., 2013).

Norepinephrine is synthesized from dopamine. In the neurons of the locus coeruleus, dopamine beta-hydroxylase converts dopamine to norepinephrine as the final obligate biosynthetic step. When dopaminergic functional output collapses at Step 2, the substrate available for norepinephrine synthesis in the locus coeruleus is correspondingly reduced. The resulting noradrenergic deficit is not an independent pathological process. It is a direct downstream chemical consequence of the dopaminergic cascade: the same functional loss that disrupts frontostriatal regulation also depletes the precursor pool on which the LC-NE system depends. This noradrenergic insufficiency flattens slow-wave sleep architecture and disrupts the infraslow oscillations that drive the glymphatic pump, providing a mechanistically complete account of why dopaminergic cascade failure leads directly to glymphatic clearance failure.

Dopaminergic dysfunction impairs slow-wave sleep through this noradrenergic pathway (Dzirasa et al., 2006). Slow-wave sleep is the physiological requisite for glymphatic system operation (Xie et al., 2013), which functions through astrocytic AQP4-mediated bulk flow during deep sleep (Mestre et al., 2020). Cerebrospinal fluid influx through the glymphatic pathway has been identified as a primary driver of acute tissue swelling following ischemic insult, demonstrating that the glymphatic system actively mediates fluid exchange and clearance (Mestre et al., 2020). As glymphatic function falls, metabolic waste and inflammatory mediators accumulate in the prefrontal cortex and basal ganglia.

Aquaporin-4 channel expression and localization in astrocytic endfeet is the primary determinant of bulk cerebrospinal fluid flow through the glymphatic compartment (Mestre, Mori, & Nedergaard, 2018). The link between sleep disruption, glymphatic impairment, and downstream neuronal stress has been established across multiple contexts, including traumatic brain injury and aging (Piantino, Lim, Newgard, & Iliff, 2019). This means that Step 3B severity is the product of two compounding factors: how severely dopaminergic dysfunction has impaired sleep quality, and how efficiently the glymphatic system can clear waste per unit of remaining sleep. Both are genetically shaped.

The Step 3B pathway creates its own feedback loop. Dopaminergic dysfunction impairs sleep. Impaired sleep halts glymphatic clearance. Uncleared metabolic waste accumulates in dopaminergic regions. The accumulation induces further cellular stress. That stress deepens the dopaminergic functional loss. Sleep quality deteriorates further. The glymphatic system falls further behind. Clinically, Step 3B manifests as nonrestorative sleep, symptoms that worsen throughout the day, post-exertional malaise with delayed recovery, and crashes after periods of activity (Graham et al., 2021).

Post-exertional malaise is the clinical signature that emerges when both Step 3 pathways are active. Physical or cognitive exertion dramatically increases metabolic waste production and excitatory demand, requiring a surge in dopaminergic regulatory support that the Step 1-compromised system cannot provide. The Step 3A pathway responds with glutamatergic overdrive and escalating metabolic stress. The Step 3B pathway cannot clear the resulting waste accumulation. The characteristic 12-to-48-hour delay in symptom collapse directly reflects the time course of metabolic waste accumulation exceeding the impaired glymphatic clearance capacity (Graham et al., 2021). This is not a psychological response to activity. It is a mechanically predictable consequence of a system where waste production from 3A overwhelms the clearance capacity of 3B.

Compensatory Homeostasis and the Delay to Collapse

The initiating trigger and the initiation of the cascade are not simultaneous. When Step 1 injury first occurs, dopamine synthesis and release begin to decline. The system does not immediately fail. The brain is a highly compensating network, and it responds to a reduction in dopaminergic output by drawing on its remaining capacity to maintain normal baseline operations. Receptor sensitivity upregulates. Compensatory firing patterns emerge. Downstream circuits sustain function by consuming capacity rather than reporting deficits. During this compensatory phase, the patient may feel no different than before the infection.

Symptoms do not emerge until compensatory capacity is exhausted. Only when the accumulated functional loss has depleted the remaining compensatory margin does dopaminergic output cross below the critical threshold, and only at that point does the Step 2 circuit failure become clinically visible. This is the mechanistic explanation for the characteristic delayed onset of Long COVID symptoms relative to the acute infection: the system was not failing after the infection resolved. It was compensating. The clinical presentation represents the point at which compensatory capacity became insufficient. The initiating trigger may have occurred days or weeks earlier.

A reduction in dopaminergic functional output only becomes clinically apparent once compensatory capacity has been exhausted. An individual with greater capacity may compensate for weeks before crossing the threshold, or may never cross it at all. An individual whose capacity was already constrained by genetic architecture or perimenopausal hormonal withdrawal may cross within days. The duration of the compensatory phase is a genetically individuated parameter, not a fixed property of the disease. This is why the delay between the initiating trigger and symptom onset varies across individuals, and why the severity of the initiating trigger alone is a poor predictor of the eventual neurological syndrome.

Supporting Evidence

Functional disruption in Neurological PASC maps specifically onto the distributions of dopaminergic pathways (Wang et al., 2025). This pattern follows directly from the architecture of the model rather than from the identity of any initiating trigger.

Advanced functional neuroimaging demonstrates disruptions in local neural activity and neurovascular coupling in COVID-19 survivors that track onto dopamine receptor and dopamine transporter distributions, with correlation to depression severity and cognitive performance (Wang et al., 2025). Large-scale structural neuroimaging in UK Biobank participants identifies selective changes in brain regions corresponding to dopaminergic networks following SARS-CoV-2 infection (Douaud et al., 2022). The spatial specificity of these disruptions is itself evidence for the cascade model: a generalized neuroinflammation model would not predict disruption patterned this way. PET-based metabolic imaging shows hypometabolism in olfactory, limbic, and prefrontal regions in Long COVID patients, a pattern consistent with dopaminergically innervated pathway disruption (Guedj et al., 2021).

Reports of Parkinsonian signs in a subset of post-COVID patients are consistent with nigrostriatal involvement (Anghelescu et al., 2022). The broader relationship between SARS-CoV-2 and dopaminergic neurodegeneration, including neuroinflammatory mechanisms mediated by microglia that directly suppress nigrostriatal function, has been extensively documented (Mancini 2023). The basal ganglia's role in regulating the motivational and fatigue dimensions of this dysfunction has been specifically characterized in the context of Long COVID (Rudroff, 2024).

SARS-CoV-2 has been shown to compromise cellular pathology in human dopaminergic neurons (L. Yang et al., 2024). Within the DTC framework, this supports the role of SARS-CoV-2 as one initiating trigger rather than defining the cascade itself. Discovery of additional initiating triggers producing the same downstream architecture would therefore corroborate, rather than compete with, the model. Commentary on this work has noted its significance for understanding how COVID-19 selectively compromises the dopaminergic system (Eidhof, Twohig, & Falk, 2024). Animal model evidence demonstrates that SARS-CoV-2 can persist in the brainstem and produce a neurodegenerative transcriptomic signature characterized by altered dopaminergic and glutamatergic synaptic gene expression, along with persistent depression-like behavior and memory impairment (Coleon et al., 2025). Post-mortem studies show perivascular accumulation of proteins and inflammatory residues in COVID-19 brains, particularly in dopaminergically innervated regions such as basal ganglia and prefrontal cortex (A. C. Yang et al., 2021; Younger, 2025).

A controlled study measuring Neurofilament Light and Glial Fibrillary Acidic Protein at a median of 69 weeks post-infection found no significant differences between Long COVID patients and recovered controls (Omdal et al., 2026). Normal NfL, normal GFAP, and the absence of diffuse neuroinflammatory signals confirm no active neurodegeneration or large-scale cell death. The brain's hardware is undamaged. What standard biomarkers cannot measure is the functional state of that hardware.

A meta-analysis of inflammatory cytokines in PASC patients found that levels of IL-6, IL-2, MCP-1/CCL2, TNF- α , IFN- γ , and IP-10/CXCL10 were significantly higher in PASC patients compared to those without the condition, with IL-6, MCP-1/CCL2, TNF- α , and IFN- γ remaining elevated across both early (<6 months) and late (6 months) time windows (Dornas et al., 2026). This persistent inflammatory signature supports the hypothesis of chronic, low-level immune activation contributing to sickness behavior in Long COVID.

Women with lower systemic inflammation, as measured by C-reactive protein, have been shown to experience steeper cognitive decline with age, suggesting that inflammatory status interacts with sex in determining cognitive outcomes (Noss, Millwood, & Kuhlman, 2022). This may have implications for understanding sex-specific vulnerabilities in Long COVID, where lower inflammatory markers could paradoxically be associated with worse cognitive trajectories.

The initiating trigger alters synaptic biochemistry—depleting cofactors such as BH4, perturbing astrocytic metabolic output, and compromising the dopaminergic support infrastructure—without killing the cells in sufficient numbers to generate a detectable plasma signal. The system is not degenerating. Pathways cannot maintain stability, excitatory environments cannot be reset, and clearance loops cannot keep pace. The cascade produces a chronic failure of regulation, not a chronic process of structural destruction. Multimodal neuroimaging studies confirm that alterations in

brain tissue microstructure and neurochemical profiles occur in Long COVID without the structural damage markers that standard plasma biomarkers would detect (**dacostaaguayo2024**; Furevik et al., 2025; Tuomaala et al., 2026).

Large-scale prospective data from the RECOVER cohort confirm that female sex is associated with elevated risk of Long COVID, an effect that is age-dependent and strongest in the 40–54 year range, consistent with perimenopausal status (Fain et al., 2025; Shah et al., 2025). Estrogen directly enhances dopamine synthesis. It upregulates the activity of dopamine-synthesizing enzymes, increases dopamine receptor expression, and modulates dopamine transporter function (Jacobs & D’Esposito, 2011; McEwen, 2002). During perimenopause, the decline and eventual withdrawal of estradiol removes this trophic support from the dopaminergic system. The result is a reduction in dopaminergic functional capacity. Perimenopausal women are operating with lower dopaminergic functional output than they were pre-menopause, and therefore closer to the functional threshold before an initiating trigger occurs.

When SARS-CoV-2 infects the astrocytic support network in a perimenopausal woman, the same initiating trigger that would leave another individual below the threshold for cascade initiation may instead initiate the DTC. Estrogen withdrawal is itself a trigger mechanism, functionally parallel to viral infection of astrocytes. Both reduce dopaminergic functional output. For women carrying additional genetic constraints at Step 1, the combination of estrogen withdrawal and astrocytic infection compound any genetic predispositions resulting in the functional loss.

Proteomic profiling has confirmed sex-specific signatures in Long COVID, with VEGFA overexpression specifically elevated in postmenopausal women (Farré et al., 2025). Immune profiling demonstrates that female and male Long COVID patients exhibit distinct immunological and endocrine profiles, with lower testosterone levels in females correlating with higher symptom burden (Silva et al., 2024).

The existing literature documents serotonin depletion in Long COVID patients, including reduced circulating serotonin in the post-acute phase and gut-brain axis serotonergic disruption during and following acute infection (Mannekote Thippaiah, Amanullah, Huang, Goldschmidt, & Pradhan, 2023). During acute infection, early inflammatory disruption and gut-brain axis dysregulation can deplete peripheral and central serotonin. However, the serotonergic system does not possess the immense, high-overhead unmyelinated axonal arbors that characterize midbrain dopaminergic projection neurons. Nor does it function as the primary inhibitory regulatory brake on frontostriatal and limbic circuit excitation. Serotonergic depletion produces bounded and largely localized dysfunction: mood dysregulation, heightened sensory sensitivity, and disrupted gastrointestinal motility. The system can compensate to a degree, and its depletion does not engage the positive feedback loops that define the cascade architecture.

The severe, multi-systemic neurological Long COVID phenotype—persistent post-exertional malaise, cognitive pathway collapse, autonomic dysregulation, and glymphatic clearance breakdown—is initiated not by serotonin depletion but by the threshold breach of dopaminergic regulatory capacity. It is the dopaminergic system’s role as the primary inhibitory modulator of frontostriatal excitation that converts its failure into a cascading, self-amplifying collapse rather than a localized deficit. When serotonin falls, the system loses a dimension of affective and homeostatic regulation. When dopamine falls below the stability threshold, the excitatory brakes are released and the cascade architecture described here is engaged.

The Polygenic Risk Architecture

Vulnerability to neurological PASC is shaped by the genetic architecture of the host at each step of the cascade independently. At Step 1, variants governing entry efficiency, astrocytic metabolic function, and dopaminergic synthesis capacity determine how deeply the initial functional breach depletes dopaminergic functional output (Asselta, Paraboschi, Mantovani, & Duga, 2020; Pecoraro, Cuccorese, & Trenti, 2023). At Step 2, variants governing innate immune reactivity determine how sustained the microglial response becomes, while variants governing receptor density, transporter expression, and G-protein signaling dynamics determine which pathways lose stability first and most severely (Eisenstein et al., 2016; Gizer et al., 2009). At Step 3A, variants governing excitatory termination, mitochondrial redox capacity, and catecholamine degradation determine the speed and depth of metabolic overload (Gizer et al., 2009; Nayeri et al., 2022). At Step 3B, variants governing glymphatic architecture and astrocytic function determine whether clearance can keep pace with production (Mestre et al., 2018).

Rather than treating this genetic architecture as a static catalog, the model requires that it function as a *representative parametric profile*: a set of genetically encoded dials that determine dopaminergic functional output, proximity to the functional threshold, and the severity of downstream propagation within a single continuous dynamical system. These dials do not operate independently. They compound. Each step constrains the system parameters that govern failure in the next.

A central requirement of this framework is that its polygenic architecture must be empirically testable. If the cascade relies on compounding vulnerabilities, the statistical distribution of these genetic variants must align with the epidemio-

logical distribution of the disease. The functional variants driving the cascade must be common polymorphisms, not rare mutations. If the vulnerabilities required rare mutations, the high global prevalence of Long COVID would contradict the model. Table 1 outlines the critical functional nodes where common polymorphisms are predicted to intersect to drive the cascade. The expected minor allele frequencies should be consistent with widespread population-level risk rather than isolated rare variant effects. An integrative multi-omics framework applying Mendelian randomization and transcriptomic analysis to Long COVID cohorts has independently identified a broader set of candidate causal genes organized around immune regulation, cell cycle control, and viral response pathways (Pinero et al., 2025).

Table 1: Representative polygenic risk loci across cascade tiers. The model predicts these functional vulnerabilities to be common polymorphisms within the general population, with minor allele frequencies consistent with the observed prevalence and heterogeneity of neurological PASC.

Step	Functional Target	Representative Loci
1	Viral entry efficiency	ACE2 variants
1	Dopamine synthesis capacity	TH, MTHFR
2	Innate immune reactivity	IL-6, TLR pathway
2	Receptor density & transport	DRD2, SLC6A3
2	G-protein signaling termination	RGS4, RGS9
3A	Excitatory termination	RGS4, RGS9
3A	Mitochondrial redox capacity	SOD2, TXNRD2
3A	Catecholamine degradation	MAOA
3B	Glymphatic clearance efficiency	AQP4

Because a significant portion of the population is expected to carry at least one allele conferring vulnerability at Step 1, the model predicts that limited reductions in dopaminergic functional output, manifesting clinically as prolonged post-viral fatigue or transient brain fog without multi-system failure, should be relatively common (hastie2022; Bell et al., 2021). This represents limited propagation through the cascade before widespread downstream dysfunction develops. However, as the cascade propagates into Step 2 and Step 3, the model relies on the principle of compounding probability. Assuming independent assortment, the statistical likelihood of an individual carrying vulnerable alleles across the synthesis, immune, receptor, excitatory, and clearance nodes simultaneously decreases with each additional step.

This compounding risk structure generates a hypothesis regarding the clinical spectrum of Long COVID. In deeply phenotyped cohorts, mild, single-domain presentations will correlate with isolated genetic vulnerabilities at Step 1 or 2. The severe, debilitating phenotype characterized by multi-system failure and profound PEM will correlate with a stacked polygenic intersection across all three steps. The extreme statistical long tail of severe neurological PASC should mathematically mirror the rarity of this full polygenic stack.

The first published GWAS of Long COVID identifies FOXP4 as a significant locus (Lammi et al., 2025), consistent with a distributed polygenic architecture rather than a single dominant signal. Combinatorial genetic analysis across independent cohorts has demonstrated that the genetic risk architecture of Long COVID is substantially broader than standard GWAS approaches reveal, with robust reproducibility across ancestrally diverse populations (Sardell et al., 2025).

Testable Predictions

The cascade model generates structural, biological, clinical, and genetic predictions that distinguish it from competing frameworks and that are accessible to empirical test. Because these predictions arise from the architecture of the cascade rather than from a specific initiating trigger, corroboration from different initiating triggers strengthens rather than weakens the framework.

Neurologically, the model predicts that functional disruption in post-COVID cohorts should map onto dopaminergic pathway distributions specifically, not onto the distributions of other neurotransmitter systems, and that this mapping should correlate with symptom severity (Wang et al., 2025).

Genetically, the model predicts that GWAS in deeply phenotyped Neurological PASC cohorts should find enrichment across the functional steps described here: innate immune regulation, dopaminergic synthesis and receptor architecture, excitatory termination, and glymphatic function, rather than a single dominant risk locus (Lammi et al., 2025; Sardell et al., 2025). The model predicts that genetic risk will be additive across steps, with individuals carrying variants at multiple levels showing disproportionate severity relative to those carrying variants at one level only.

Epidemiologically, the model predicts that severity of neurological PASC will correlate negatively with acute infection severity in infected individuals because the most genetically vulnerable are those who were already near the cascade threshold before exposure, not those who mounted the largest acute immune response (Vanichkachorn et al., 2021).

Interventionally, the model predicts that symptom-targeted treatments addressing downstream outputs of the cascade, such as anti-inflammatories and symptom management, will show limited efficacy in individuals with deep polygenic risk at multiple steps because the primary pathology is failure of a regulatory architecture rather than an isolated downstream symptom (Knopman et al., 2026). Interventions targeting dopaminergic synthesis support, excitatory dampening at Step 3A, and sleep architecture restoration to enable Step 3B clearance represent causally motivated therapeutic strategies.

The parametric dial model generates three additional predictions whose specificity distinguishes them from general cascade claims. These are prospective predictions derived from the dial architecture, not post-hoc accommodations to observed data.

Prediction A (Baseline Depletion Kinetics). Under identical acute inflammatory stress, individuals carrying the stacked *IL-6* rs1800795 G/G and *MTHFR* rs1801133 G/A cofactor restriction will show a significantly faster decline in cerebrospinal fluid L-DOPA and BH4 levels compared to wild-type controls, detectable before any overt cognitive symptoms appear. The Step 1 baseline dials set the depletion rate. A null result—equivalent depletion rates across genotype groups—would falsify the parametric architecture at Step 1. (This claim draws on the genetic evidence described in the Polygenic Architecture section, particularly the *IL-6* and inflammatory gene variants discussed above.)

Prediction B (Compensatory Latency and Synaptic Buffer Capacity). The temporal delay between the initiating trigger and the onset of clinical symptoms is a function of synaptic buffering capacity as genetically determined, and is the measurable empirical signature of the compensatory phase. Individuals with lower D2 receptor density (*DRD2* rs1800497 G/A) and rapid enzymatic dopamine clearance via fast-activity *COMT* variants will possess a smaller compensatory buffer: the capacity is both shallower and depleted faster, meaning the functional threshold is crossed sooner following equivalent Step 1 injury (Felten, Montag, Markett, Walter, & Reuter, 2011). Individuals with robust receptor architecture and slow *COMT* clearance will sustain a longer compensatory phase, exhibit later symptom onset, and express a more gradual threshold-crossing profile. The model predicts that this delay-to-onset will be a continuous function of the combined synaptic buffer genotype, measurable prospectively in cohorts with timestamped symptom onset relative to confirmed infection date, independent of acute illness severity. A null result—equivalent delay-to-onset across genotype groups—would falsify the compensatory buffer architecture at Step 2.

Prediction C (PEM Recovery Non-Linearity). The duration of a post-exertional malaise crash will scale non-linearly with the interaction of *GRIN1* rs1126442 G/A (which governs excitatory waste generation rate at Step 3A) and *AQP4* rs2075575/rs162009 G/A (which governs glymphatic drainage efficiency at Step 3B) (Kress, Iliff, Xia, et al., 2014). The model specifically predicts a synergistic double-hit effect: carriers of both risk variants will experience exponentially longer PEM recovery windows than carriers of either variant alone, because the two dials operate on different sides of the same metabolic bottleneck. A linear additive relationship between the two genotype groups would be inconsistent with the cascade model's prediction of compounding non-linearity at Step 3.

Conclusion

Neurological PASC has resisted causal explanation because the field has largely focused on what SARS-CoV-2 does to the brain rather than on how the nervous system behaves once dopaminergic functional output can no longer sustain the systems that depend upon it. The Dopaminergic Trophic Cascade provides that systems-level explanation. The object of this hypothesis is not the initiating trigger, but the architecture that transforms an initiating trigger into a coherent neurological syndrome. Accordingly, the cascade is defined independently of the initiating trigger and applies wherever dopaminergic functional output is reduced sufficiently to initiate the cascade.

The initiating trigger reduces dopaminergic functional output. Whether that reduction is sufficient to initiate the cascade depends on the host architecture. Which pathways fail first depends on the genetic architecture of receptor density and signaling dynamics. How rapidly downstream dysfunction develops depends on the genetic architecture governing excitatory regulation, mitochondrial resilience, and glymphatic efficiency. At every level, the host architecture shapes the cascade. The initiating trigger starts the process; the architecture determines how it unfolds.

This framework explains why the symptoms consistently co-occur as a syndrome, why they are delayed, why exertion collapses the system, why anti-inflammatory interventions often fail to resolve the neurological syndrome, why biomarkers of structural damage are absent in the chronic phase, and why certain individuals—those already operating closer to the functional threshold because of their genetic architecture, hormonal environment, or other factors affecting dopaminergic functional output—are disproportionately vulnerable. It explains why a mild acute infection can produce a

devastating chronic course. It explains why recovery is not merely slow, but may be structurally prevented in individuals whose genetic architecture has closed the clearance loop.

This framework also generates predictions beyond Neurological PASC. If the Dopaminergic Trophic Cascade is a general systems failure mode, then other initiating triggers that reduce dopaminergic functional output sufficiently should be capable of initiating the same cascade. Post-viral syndromes such as ME/CFS, despite differing initiating triggers, may exhibit overlapping downstream neurological syndromes because the regulatory architecture being disrupted is the same (Takamatsu & Kuga, 2025). Altered TRPM3-mediated cytosolic calcium influx has been shown to significantly impact mitochondrial calcium mobilization in natural killer cells of ME/CFS patients, suggesting that mitochondrial dysfunction may be a shared mechanism across post-viral syndromes (Magawa, Eaton-Fitch, Muraki, & Marshall-Gradisnik, 2026). This places Long COVID and certain forms of Parkinsonism on a continuum of dopaminergic vulnerability—not as identical diseases, but as outcomes of the same systems failure mode triggered by different events in hosts with different baseline functional output.

The framework also reframes the question of treatment. The cascade model points toward interventions that restore dopaminergic synthesis support, dampen excitatory overload at Step 3A, or structurally restore slow-wave sleep to re-enable Step 3B clearance. None of these are novel drug targets. They are existing mechanisms, now motivated by a framework that explains why they are the right ones.

By integrating neuroimaging findings, circulating biomarker data, polygenic risk architecture, and demographic hormonal dynamics, this framework provides a comprehensive, testable, and parsimonious account of neurological PASC and a foundation for interventions aimed not at the symptoms, but at the system that has lost the capacity to regulate them.

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